Smart AI Fan: Round 2 Project Status

TECHNOMANCERS (T-2026) | atomQUEST 2026, Track 1 Fan | Aditya Institute of Technology and Management, Tekkali, Dept. of ECE

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Status date: 5 November 2026 | Accompanies the submitted walkthrough video (about 1 min 29 s) | Repository: github.com/harsha-vardhan-2006/atomquest

1. Summary

The Round 1 concept (a 1,400 mm ceiling fan with a two-axis tilt head, sensor-based comfort control and a safety supervisor) now has a complete native SolidWorks design package: 18 parts, 3 assemblies, 18 STEP exports, 7 printable STL meshes and 21 drawing sheets, with the mass, torque, load and clearance calculations behind them. The original motor, rotor, shaft and bearings are retained and shown only as a reference envelope. The new hardware is a ceiling plate, downrod, stationary yoke, outer tilt ring, inner cradle, servo brackets, pivots, shroud and a three-blade rotor.

Update in this version: the tilt servo is selected and its cost is closed out. Four DSSERVO DS3235 (35 kg-cm class) servos, two per axis, cover the 4.5 N.m per-axis torque need with margin and fit the 20 x 40 mm mount in the CAD. Bought as a single 10-piece lot from the manufacturer tier, the four fitted servos cost about INR 3,600 to 4,600 instead of INR 7,200 at retail, and the six spare servos cover the lever-arm pull tests that section 5 requires. The servo line therefore stays inside the INR 3,500 to 6,500 that the full-fan estimate already carried, and the full-fan estimate does not rise.

Every item below is marked Completed or Pending. Pending items are the remaining engineering work before any hardware is built. No powered operation is approved.

2. Work status

Item	Status	Detail
Native SolidWorks parts	Completed	18 parts, each one solid body, reopened and rebuilt through the API.
Assemblies	Completed	FD_ROTOR (4 components), FD_TILT (11), FD_SMART_FAN_ASSEMBLY (7); blades placed at 0/120/240 deg.
STEP and STL exports	Completed	18 STEP and 7 STL files; 17 of 18 STEP files match the native parts within 0.5 mm after re-import.
Drawing sheets	Completed	21 sheets with third-angle views and 172 dimension entities.
Engineering calculations	Completed	Mass, inertia, tilt torque, blade root loads, ceiling anchor loads and tilt clearance (analytical estimates).
Tilt motion study	Completed	45 poses checked with interference detection. It found overlap between the tilt group and the top shroud, so geometry rework is listed below.
Servo selection	Completed	4 x DSSERVO DS3235, 35 kg-cm class, run at 7.4 V (2 per axis). Stall torque 6.9 N.m per axis against the 4.45 N.m (X) and 3.62 N.m (Y) need; 40 x 20 x 38.5 mm body fits the 20 x 40 mm mount. Bought as a 10-piece lot at about INR 700 to 880 per servo (about INR 900 to 1,150 landed), so the four fitted servos cost about INR 3,600 to 4,600 and the six remaining are spares for the pull tests. Details and alternatives in section 5.
Safety and test plan	Completed	Compliance gap matrix and a 9-stage test plan documented.
Walkthrough video	Completed	Screen recording of the three assemblies plus a short clip of the original fan hardware.
Tilt geometry rework	Pending	Open up the shroud or lower the tilt group, move the travel stop, clear 7 rest-pose overlaps, then re-run the motion study (target travel +/-20 deg and +/-15 deg).
Mass reduction	Pending	Moving mass is 10.37 kg against the 6.80 kg budget; blade is 0.330 kg against the 0.165 kg target. Hollow or rib the blade and lighten the yoke and cradle.
Servo torque confirmation	Pending	Re-check the servo margin once the original motor is weighed. Bench-test one servo on a lever arm, and extend the CAD from 2 brackets and 1 horn to 4 servos with 25T horns.
Original motor measurement	Pending	Motor envelope, mounting PCD, shaft and bearings are assumed values until measured.
Assembly mate clean-up	Pending	Repair or suppress the failed mates shown by error markers in the FD_TILT and FD_ROTOR trees, then re-record the video if needed.
Drawing completion	Pending	GD&T, fit tolerances, title block and BOM table.
Material assignment	Pending	Assign SolidWorks materials so CAD mass matches the calculation sheet.
Print and fit test	Pending	Print small fit-check parts (shroud, cover, brackets, collar) and check the unpowered tilt by hand.
Aerodynamics and containment	Pending	No CFD or test yet; rotor containment (130 J basis) is unconfirmed. Powered running stays unapproved until these are done.

3. Key figures

Parameter	Current value	Target / basis	Status
Blade tip speed at 380 RPM	27.86 m/s	30 m/s cap	Completed
Ceiling anchor load (4 anchors)	122 N static, 367 N dynamic, 980 N abuse	Anchor rating to be confirmed	Completed
Blade centrifugal load at 380 RPM	190 N, 0.36 N.m root bending	Blade strength test	Completed
Blade-tip to ceiling at 20 deg tilt	112 mm	Hard stop at 27 deg	Completed
Servo torque per axis	6.9 N.m stall (2 x DS3235 at 7.4 V)	Need 1.4 to 4.5 N.m (two calculation sheets)	Completed
Tilt servo cost, 4 fitted	INR 3,600 to 4,600 (bulk lot of 10)	INR 3,500 to 6,500 actuator line	Completed
Moving mass	10.37 kg (total 12.49 kg)	6.80 kg	Pending
Rotor energy at 380 RPM	69 J (calculation) / 77 J (CAD)	130 J containment basis	Pending
Tilt travel	Rework in progress	+/-20 deg and +/-15 deg	Pending

Two scripts estimate mass and inertia with different groupings, which is why two energy values and two servo-torque figures are shown; they will be reconciled after the motor is weighed.

4. What the video shows

The recording (1280 x 720, about 89 s) rotates and inspects FD_SMART_FAN_ASSEMBLY, then FD_ROTOR (hub with three blades) and FD_TILT (yoke, outer ring, inner cradle, motor adapter and cover, bushings, servo brackets, travel stop), and returns to the full assembly. Short clips show the original fan hardware (motor housing and wiring), the part the design keeps.

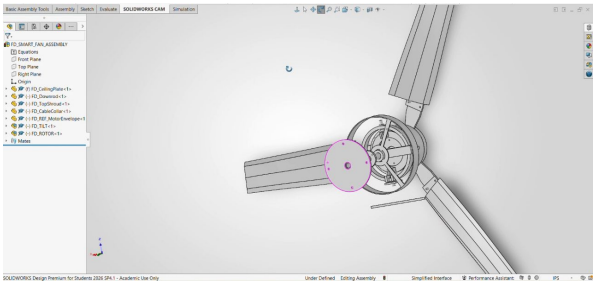


Fig. 1: FD_SMART_FAN_ASSEMBLY (video frame)

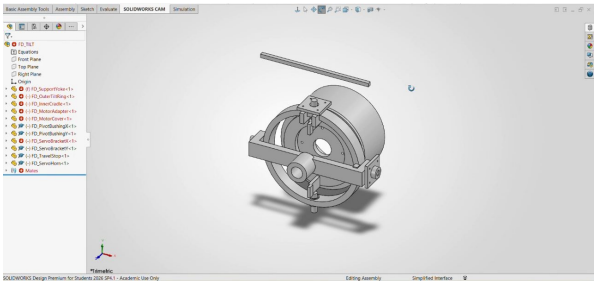


Fig. 2: FD_TILT assembly (video frame)

5. Servo selection (Completed)

Why four servos. REQ-20 asks for four servos (X1+X2 on one axis, Y1+Y2 on the perpendicular axis) and REQ-35 asks for the actuator torque to be recalculated for that layout. The torque need comes from the repository: 4.45 N.m (X) and 3.62 N.m (Y) per axis in MASS_AND_TORQUE_REPORT.md (includes a 2.0 design margin, 0.85 linkage efficiency and a placeholder motor mass), and 1.41 / 1.45 N.m in the second calculation sheet. The servo mount in the CAD is a standard 20 x 40 mm servo footprint.

Selected: 4 x DSSERVO DS3235 (35 kg-cm class coreless digital servo, metal gears, IP66), powered at 7.4 V, two servos per axis.

DS3235 datasheet item	Value (manufacturer datasheet)
Size / weight	40 x 20 x 38.5 mm, 60 g, 25T output spline, double ball bearing, 300 mm lead
Supply voltage	5 to 7.4 V; operating temperature -15 to 70 C
Stall torque at 5 / 6 / 7.4 V	29 / 32 / 35 kg-cm = 2.84 / 3.14 / 3.43 N.m
Stall current at 5 / 6 / 7.4 V	1.9 / 2.1 / 2.3 A (idle 5 mA)
Speed at 5 / 6 / 7.4 V	0.13 / 0.12 / 0.11 s per 60 deg (no load)
Control	PWM 500 to 2500 us, 1500 us neutral, 50 to 330 Hz, 180 or 270 deg travel, dead band 2 us

How it meets the requirements (7.4 V; need = repository figure; "nominal" = the need divided by its 2.0 design margin: X 2.23, Y 1.81 N.m).

Check	Result	Margin X / Y	Verdict
Number of servos (REQ-20)	4 servos, 2 per axis	-	Met
Both servos on an axis, stall torque	6.86 N.m vs 4.45 (X) and 3.62 (Y)	1.54x / 1.90x	Met
One servo failed, no runaway (REQ-35)	3.43 N.m vs nominal 2.23 (X) and 1.81 (Y)	1.54x / 1.90x	Met
Mount fit	40 x 20 x 38.5 mm body in the 20 x 40 mm mount	-	Met
Continuous holding, 2 servos	2.31 N.m (vendor rated torque 11.8 kg-cm each) vs nominal 2.23 / 1.81	1.04x / 1.28x	Thin on X; confirm after weighing the motor

The continuous-torque row uses a rated-torque figure from one seller listing, not the manufacturer datasheet, so treat it as indicative. If the weighed motor makes the X axis too heavy, add a gear reduction or a counterweight rather than a bigger servo. At 6.0 V the X margin falls below 1 on that basis, so keep the rail at 7.4 V and not above it.

Alternatives considered

INR from Indian shop listings found on 5 Nov 2026; listings, not quotations.

Option	Stall torque per servo	Cost, 4 pieces	Assessment
DS3225MG (25 kg-cm class)	2.40 N.m at 6.8 V (24.5 kg-cm)	INR 3,756 (INR 939 each)	Fits the old INR 3,500 to 6,500 actuator line, but stall margin is only 1.08x on X and continuous torque is below the need. Fallback only if the motor weighs little.
DS3235 (selected)	3.43 N.m at 7.4 V (35 kg-cm)	INR 3,600 to 4,600 fitted (10-piece lot); INR 7,200 at retail	Meets every row above; same footprint as the CAD mount; seven checks pass without gearing.
DS3235SSG (stainless gear)	3.43 N.m class (seller lists 34 kg-cm)	INR 9,240 (INR 2,310 each)	Same torque class with stronger gears; optional upgrade for the final fan.
MG996R (hobby class)	1.08 N.m at 6 V (11 kg-cm)	about INR 1,600	Rejected: two per axis is still below the nominal X torque, so it cannot meet the need at any price.
Futaba BLA21-12R3-C01 (industrial)	4.90 N.m max	not listed; import	Rejected: torque is adequate but it is one servo per axis, which fails the 4-servo REQ-20 layout; import cost and lead time are outside the line.
Futaba BLA21A series (industrial)	4.51 to 4.90 N.m, IP67	not listed; import	Rejected for the same layout and sourcing reason; IP67 is not needed indoors.
Festo EMMT-AS-80-M-HS-RMB	2.60 N.m stall (6.4 N.m peak)	not listed; industrial	Rejected: stall is below the need, the peak is transient only, and it needs an industrial drive on a 680 V DC bus.

The DS3235 wins on three counts. It is the cheapest part that passes every check, so the cheaper DS3225MG and the hobby-class MG996R are ruled out on torque rather than cost. It survives one servo failing, which the single-servo industrial options cannot offer because they have no second actuator on the axis. And it drops into the bracket already modelled in the CAD with a 25T spline, on ESP32 hardware PWM with no separate drive. The Futaba and Festo units solve a positioning-accuracy problem this fan does not have, at many times the price and on an import lead time.

Sourcing and cost

Sourcing tier	Quantity	Unit price	4 fitted servos	Cash out
Retail, one-off	1 each	INR 1,406 to 2,299	INR 5,624 to 9,196	same
Retail, in stock on 5 Nov	1 each	INR 1,800	INR 7,200	INR 7,200
Bulk lot (selected)	10	INR 700 to 880 ex-works; about INR 900 to 1,150 landed	INR 3,600 to 4,600 landed	INR 9,000 to 11,500 landed

The manufacturer tier is where the price actually falls: about INR 700 to 880 per servo ex-works instead of INR 1,800 at retail, roughly 55 per cent lower. Two points keep that honest. First, no MOQ-10 listing sells four pieces, so the lot has to be ten and cash out is INR 9,000 to 11,500 even though four servos go into the fan. Second, the other six are not padding: section 5 requires each servo to be pull-tested on a known lever arm before fitting, and that test destroys samples, so six covers the bench test plus one spare set. Landed figures add freight and 18 per cent GST at INR 88 to the US dollar; prices move with the exchange rate and must be confirmed with the seller before ordering.

Budget effect. The full-fan estimate carried INR 3,500 to 6,500 for the tilt actuator set. The four fitted DS3235 come to INR 3,600 to 4,600 landed, which sits inside that line, so the full-fan estimate stays at about INR 25,000 including 10 per cent contingency instead of rising to INR 28,000 to 29,600 on the retail-one-off basis. The 7.4 V supply is not priced here.

Integration notes

(1) Power the four servos from a dedicated, fused, regulated 7.4 V rail (buck from the 24 V supply) sized for 4 x 2.3 A = 9.2 A worst case; do not draw it from the ESP32 board. (2) Drive from ESP32 hardware PWM; firmware limit +/-20 deg (REQ-12), mechanical stop at 27 deg (REQ-13), rotor stopped before any tilt (REQ-11), and the safety supervisor can cut servo power. (3) The CAD still holds 2 servo brackets and 1 generic horn; add the other two brackets and 25T horns. (4) Seller specs differ for the same model (for example DS3225 listings range from 21 to 28 kg-cm), so buy against the datasheet torque and pull-test each servo on a known lever arm before fitting.

6. Cost basis and next steps

Cost (planning estimates, not quotations; docs/cost_basis.md): Round 2 Level 1 control demonstrator about INR 10,700 (7,040-14,300); one full-fan prototype about INR 25,000 (17,160-32,890), unchanged by the servo update because the selected 10-piece lot lands inside the actuator line it replaces. Tooling, test equipment, certification and shipping are excluded.

Next steps: (1) repair tilt geometry and mates, then re-run the motion study; (2) weigh and measure the original motor, then re-check the servo margin; (3) cut mass and re-run the calculations; (4) order the 10-piece DS3235 lot and a 7.4 V supply,

bench-test one axis with two servos, then extend the CAD to four servos; (5) print fit-check parts and test the unpowered tilt by hand; (6) complete drawings and materials.

7. Where to find things

Content	Location in the repository
Status report, servo decision and sizing	servo/ (this report, SERVO_SELECTION.md, servo_sizing.csv, generate_status_report.py)
Final design report, limitations, changelog	competition/atomQUEST2026/round2/FINAL_DESIGN/ (FINAL_DESIGN_REPORT.md, KNOWN_LIMITATIONS.md, CHANGELOG.md)
Parts, assemblies, STEP, STL, drawings	01_NATIVE_PARTS, 04_ASSEMBLIES, 02_STEP_EXPORTS, 03_PRINT_STL, 06_DRAWINGS (inside FINAL_DESIGN)
Calculations and reports	07_ENGINEERING_CALCULATIONS (servo_sizing.csv, mass_and_torque.csv), 08_VERIFICATION_REPORTS, MASS_AND_TORQUE_REPORT.md
Safety and test plan	FINAL_DESIGN/11_TEST_PLAN/SAFETY_AND_TEST_PLAN.md
Round 1 evidence and cost basis	docs/, data/, calculations/, submission/ at the repository root

8. Sources for the servo research

DS3235 manufacturer datasheet: hajim.rochester.edu/me/sites/kelley/me240/DS3235-270_datasheet.pdf. Indian listings and prices: rees52.com (DS3235 INR 1,919; DS3235MG INR 1,800; DS3235SSG INR 2,310; DS3225MG INR 939), roboticsdna.in (DS3235MG INR 1,406 incl. GST, out of stock), probots.co.in (DS3225 INR 2,499 incl. GST, out of stock at the time). Bulk tier: Alibaba listings for Dsservo DS3235 35KG at USD 8 to 10 per piece, MOQ 10, from Dongguan Dsservo Technology, Dongguan Sanshu Electric and Shenzhen Fengfei Electronics, ex-works. DS3225 specification: bzbuas.com and racer.lt listings. Torque conversion: 1 kg-cm = 0.0980665 N.m. Torque need: repository files MASS_AND_TORQUE_REPORT.md, servo_sizing.csv and calculations_summary.json. Futaba BLA21-12R3-C01 and BLA21A series: futaba.co.jp industrial servo pages and futabairc.com product introduction. Festo EMMT-AS-80-M-HS-RMB: Festo datasheet 5255440. MG996R: handsontec.com MG996R.pdf. Prices change; confirm with the seller before ordering.